Sultant Of Oman
Ministry of Health
Royal Hospital

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Document Management System

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The Royal Hospital
Department of Blood Transfusion Services

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Title: Blood ordering for emergency and massive transfusion protocol

Introduction:

Provision of blood components during emergencies has many challenges including risks of incorrect identification of patient, possibility of wrong blood in tube and time limitation for testing. Therefore, having a clear protocol that reduces risks and improves turnaround time is crucial. Documentation of every step during the response to the emergency is important, to be able to monitor errors and apply correction actions for future events.

1. Any emergency blood requests must be communicated immediately to blood bank at any time to Ext 9436 or 9711.
2. Blood bank must be informed clearly about the case and the requirement of the patient, blood bank must be given the following information:

- Patient name
- Hospital number
- Age and gender
- Location of the emergency
- Name/staff number of the caller.
- Number of uncross matched red cell units required.
- The likelihood of requiring further blood components.

3. The caller must send the porter immediately with patient sticker generated at the site of the emergency and send blood samples for compatibility testing when possible and as soon as possible.

4. Direct release of blood in emergency form (un x match blood Form) must be used when uncross matched blood units are issued. This form will be released with the blood and must be signed by the physician requesting the blood and returned to the blood bank.

6. Allabnormal results will be reported to the physician immediately.The blood bank will recall all red cell units found to be incompatible on retrospective testing, and request returning them to the blood bank if they are still unused and if the units are already transfused then the patient should be monitored for transfusion haemolytic transfusion (haematology team can be consulted to advise when needed).

Issuing of red cell units:

- If patient's blood group is unknown or full patient identification is unavailable; O Rh (D) negative blood will be released for a female patient of child baring age, and O Rh (D) positive blood will be released for a male patient.
- In case of known blood group with more than one historical record specific or compatible blood group products will be released.
- In case of known blood group with only one historical record O Rh (D) negative blood will be released for a female patient of child baring age, and O Rh (D) positive blood will be released for a male patient, unless a sample for blood group is sent.
- If more than 8 red cell units are issued to a particular patient and physician anticipate additional units will be required massive transfusion protocol must be activated by contacting the blood bank to activate massive transfusion protocol.
- Once massive transfusion protocol is initiated then the Laboratory Consultant Haematologist / Senior Specialist on duty will be notified.

Massive transfusion protocol activation:

- The blood bank technical staff, once massive transfusion protocol is activated by the treating medical team, will:
- Check inventory and contact CBB if the inventory will not cover the required blood components
- Start preparing the packs as per the massive transfusion protocol
- Once Samples are received, retrospective compatibility testing will always be performed in ensure appropriate action and facilitate retrieval of incompatible blood units prior to transfusion.
- If massive transfusion is initiated, please refer to appendix below (Massive transfusion protocol). MTP Pack will need to be issued in this case (4 units PRBCs, 4 units FFP, 4 units of platelets) **alternating with** (4 units PRBCs, 4 units FFP, 10 units of cryoprecipitate).

Massive transfusion protocol:

Definition of massive blood loss:

- Loss of 1 blood volume *(approximately 70ml/kg i.e., 5 L)* in 24 hours or 150 ml/hr or 50% of blood volume in 3 hours

Role of the physician in-charge:

- It is the responsibility of the physician on site to activate the MTP if the clinical situation is consistent with a very likely hood of massive transfusion emergency
- MTP is activated by calling the blood bank [EXT. 9436 or 9711(after working hours)]
- The use of rFVIIa should be discussed between treating consultant and haematology consultant

Role of the Blood Bank:

- Supply initial requested units of Packed Red cells products (PRBCs)

per MTP pack.
- If MTP is not activated and further PRBCs (>8 units) are requested within 24 hours then suggest MTP activation, document the medical officer's name & location. Notify the on-call haematologist.

MTP PACK (Table 1)**:**

- (4 units PRBCs, 4 units FFP, 4 units of platelets)**alternating with** (4 units PRBCs, 4 units FFP, 10 units of cryoprecipitate)
- However, additional units may be needed if:

- Platelet's count < 50X109/l or < 100X109/l in head injury
- Fibrinogen < 1.0 gm/l (cryoprecipitate should be used as recommended)

Investigations (Table 1)**:**

- CBC, UE1, LFT, Coagulation profile, ABG, Group and cross match.
- Should be repeated during the resuscitation, at least hourly or as per the treating team discretion, to ensure optimal replacement of blood components. However, repeat cross match is not required once greater than one volume of blood is transfused in a bleeding patient. Cross match will need to be repeated after 24 hours from commencing the transfusion.

Resuscitative end points (Table 1)**:**

- Normal PT, APTT
- Fibrinogen >1.0 gm/L
- Platelets > $50X10^9$/l or ≥ $100X10^9$/l in head injury
- PH 7.35-7.45
- Core temperature > 35.5 C
- Base deficit < - 3.0

Poor prognostic values:

- Systolic blood pressure < 70
- Temperature < 34 C
- Base deficit > - 6.0
- PH < 7.1

Recombinant F VIIa (NOVOSEVEN);

Please refer the rFVIIa administration protocol:

- Should be considered when all of the following factors are met :

1. Persistent bleeding despite adequate replacement of blood components as per the last laboratory results.
2. The cause of bleeding is not surgically correctable.
3. The patient has a reasonable prospect of survival once bleeding is controlled.
4. The patient is not hypothermic or acidotic.

- Please be aware that the use of rFVIIa in this situation will be off label
- Approval of use should be discussed at a consultant level.
- Should be prescribed by consultant and if afterhours, Verbal approval is acceptable until proper documentation is done the following day.
- For doses and administration, please refer to rFVIIa administration protocol.

Goal	Procedure	Comment
Restore circulating volume	Insert wide bore peripheral or central cannulae Give pre-warmed crystalloid or colloid as needed Avoid hypotension or urine output <0.5 ml/kg/h	14 gauge Monitor CVP Keep patient warm Concealed blood loss is often underestimated
Contact key personnel	Clinician in charge Consultant intensivist Blood bank scientist Haematologist	A named senior person must take responsibility for communication & documentation Arrange ICU bed
Arrest bleeding	Early surgical or obstetric intervention Interventional radiology	
Request laboratory investigations	CBC, PT, APTT, Thrombin time, Fibrinogen ; blood bank sample, biochemical profile, blood gases and pulse oximetry Ensure correct sample identification Repeat tests after blood component infusion	Results may be affected by colloid infusion May need to give components before results available
Maintain Hb >8 g/dl	Assess degree of urgency	Collection of spilt blood can be set up in <10 min

	Give red cells : **Group O Rh D negative**; in extreme emergency & until ABO and Rh D groups known **ABO group specific** when blood group known **Fully compatible blood** if time is permitting Use blood warmer and/or rapid infusion device if flow rate >50 ml/kg/h in adult	postmenopausal female Further serological cross match not required after 1 blood volume replacement Transfusion laboratory will complete cross match after issue
Maintain platelet count >75 $X10^9/l$	Allow for delivery time from blood centre Anticipate platelet count <50 $X10\ ^9/l$. after 2 X blood volume replacement	Allows margin of safety to ensure platelet count >50 $X10\ ^9/l$ Keep platelet count >100 $X10\ ^{9.}/l$ if multiple or CNS trauma or if platelet function is abnormal
Maintain PT & APTT < 1.5 X mean control	Give FFP 12–15 ml/kg (1 L or four units for an adult) guided by tests Anticipate need for FFP after 1–1.5 X blood volume replacement Allow for 30 min thawing time	PT/APTT >1.5 X mean normal value correlates with increased microvascular bleeding Keep ionized Ca2+ > 1.13 mmol/l

Maintain Fibrinogen > 1.0 g/l	If not corrected by FFP give cryoprecipitate (1 units for every 10 kg body weight) Should be available on-site. Allow for 30 min thawing time	Cryoprecipitate rarely needed except in DIC
Avoid DIC	Treat underlying cause (shock, hypothermia, acidosis)	Although rare, mortality is high

References:

- British Committee for Standards in Haematology: Guidelines on the management of massive blood loss. British Journal of Haematology 2006, 135, 634–641
- Sajjan Mittal and Henry G. Watson: A critical appraisal of the use of recombinant factor VIIa in acquired bleeding conditions. British Journal of Haematology 2006, 133, 355–363

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